

EchoPulse AI: A Hybrid Local Data Architecture for Multimodal Music Recommendation Integrating WASABI, MSD, and TalkPlay Datasets

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Abstract. Music recommendation has a problem that bugs me personally: emerging artists barely show up in suggestions. It’s not about their music quality. Most algorithms just chase popularity signals, creating loops that keep attention on already-famous hits. This paper introduces EchoPulse AI, a locally-deployed multimodal recommender we built to tackle this visibility gap. The system processes lyrical, acoustic, meta-data, and conversational data through a Triple-Encoder Transformer architecture, pulling together heterogeneous datasets (WASABI, Million Song Dataset, Lyrics 1950-2019, and TalkPlayData-2) via a five-stage local pipeline. We went with local deployment instead of cloud infrastructure because of cost constraints and internet instability—pretty common around here in Popayán, Colombia. Maybe the toughest part was unifying the Master Table, synthesizing info from multiple sources into a curated dataset of 5,710 tracks with complete multimodal coverage. Experimental evaluation against collaborative filtering, BERT4Rec, and SASRec baselines gave us results that honestly exceeded what we expected. EchoPulse AI hits competitive accuracy ($\text{Hit@10} = 0.312$) while boosting Intra-List Diversity by 45.3% and, most importantly, Emerging Artist Discovery by 157.9%. Core retrieval components run at sub-250ms latency on consumer hardware. The current setup has some clear limitations though: it mostly recommends English-language content, needs adaptation for Spanish-language music, and requires more work to become adaptable to arbitrary music databases. Scaling to industrial catalogs and running online user studies are key next steps.

Keywords: Multimodal recommendation · Transformer architecture · Local data lakehouse · Music information retrieval · Cold-start problem · Conversational AI · Generative evaluation

1 Introduction

Digital streaming totally reshaped the music business. Global revenues hit USD 29.6 billion in 2024, showing massive growth in the streaming era [37]. But for independent musicians, this economic boom often feels like an illusion. The first author spent 14 years playing drums in various bands across Colombia, and the reality on the ground is pretty stark: new artists just don't get recommended. Not because their music lacks quality. Because algorithms systematically favor established acts.

Most platforms still lean heavily on collaborative filtering. They just look at what people already click on. Works fine for global superstars, but creates a brutal cold-start problem for anyone else [24, 36]. If a new track has no play history, the algorithm ignores it. Creates this vicious feedback loop. In Latin American music scenes, the situation's even worse. Local artists aren't just competing with global hits; they're drowning in a sea of algorithmically-generated background music.

Content-based approaches offer a way out. By looking at actual lyrics, acoustic properties, and metadata of a song, it becomes possible to recommend based on what the music actually sounds like [35]. Comprehensive treatments of recommender systems theory [34] provide the foundation where modern Transformer-based approaches are built. The seminal work "Attention Is All You Need" by Vaswani et al. [1] particularly inspired this project. Its ability to map long-range dependencies makes Transformers ideal for capturing the complex structure of musical content [40].

This paper introduces EchoPulse AI. We built it as a locally-deployed multi-modal recommender within the IMS Research Group at Fundación Universitaria de Popayán. Why local? Simple answer. The research group had a limited budget for cloud services, and internet connectivity in the region is notoriously unstable. Instead of seeing these constraints as a weakness, the team used them to force a design that prioritizes efficiency, reproducibility, and zero cloud dependency.

Three main contributions emerge from this work. First, a hybrid local data architecture is proposed. It follows a five-stage paradigm to merge messy, heterogeneous music datasets without needing AWS or GCP. Second, a Triple-Encoder Transformer is designed. It processes lyrics, audio, and conversational data through specialized modules, backed by strict mathematical formulations. Third, an end-to-end evaluation is provided. Results prove that you can drastically improve diversity and surface emerging artists without tanking accuracy metrics.

Building the Master Table was easily the hardest part of this project. The team had to reconcile completely different ID systems, handle massive amounts of missing data, and force four distinct datasets into a single, clean collection of 5,710 tracks. That data engineering struggle directly shaped the architecture described in the following sections.

The rest of the paper walks through the approach step-by-step. Section 2 covers the theoretical framework, background, and research hypotheses. Section 3 dives into the five-stage pipeline and model architecture. Hardware setup and metrics are detailed in Section 4, followed by an honest look at limitations in

Section 5. Finally, Section 6 wraps up findings and what they mean for the future of equitable music recommendation.

2 Framework

2.1 Theoretical Framework

Attention and Transformer Architecture Back in 2017, Vaswani and colleagues dropped a paper that changed everything. Their Transformer architecture [1] introduced this clever self-attention mechanism:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (1)$$

The math looks intimidating at first glance, but the idea’s straightforward. You’ve got query, key, and value matrices ($\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$), and the model figures out which parts of the sequence matter most. This works surprisingly well for music because songs have structure—verses, choruses, bridges—that transformers can pick up on. Comprehensive treatments of recommender systems theory [34, 35] provide the foundation where modern Transformer-based approaches are built.

Multimodal Fusion Principles You can’t just look at one aspect of a song. Lyrics matter, acoustic features matter, metadata matters. Baltrušaitis et al. [40] laid out theoretical groundwork for combining multiple data sources. For fusing audio with text, Porcaro et al. [12] explored some techniques. Cross-modal attention mechanisms showed up in vision-language work like ViLBERT [18] and LXMERT [19], and we adapted those theoretical concepts for cross-attention fusion in music. For the emotional side, Russell’s circumplex model [42] and Eerola’s review [30] help map acoustic and lyrical features to emotional dimensions.

Local Data Architecture Theory The Lakehouse paradigm is the new hotness in data management. Armbrust et al. [15] showed how you get data lake flexibility with data warehouse transactional capabilities. Technologies like Delta Lake, Apache Iceberg, and Apache Hudi handle ACID transactions over object storage. Pimentel et al. [45] discussed data versioning approaches that informed our design. These theories allow for reproducibility and traceability, which we adapted locally using parquet-based storage.

Conversational AI and RAG Retrieval-augmented generation (RAG) frameworks [16, 17] provided methodological foundation for integrating external knowledge with generative models. This theoretical connection between user intent and structured retrieval is exactly what we needed for connecting natural language to specific songs.

2.2 Background

Sequential Recommendation Models BERT4Rec [4] took Transformers further with bidirectional encoding. Improved accuracy, sure, but also inherited all the popularity bias from training data. SASRec [3] went a different route with self-attention for sequential recommendation. Both became our baseline comparisons. Most platforms still lean heavily on collaborative filtering, creating a brutal cold-start problem for new tracks [24, 36].

Music Information Retrieval (MIR) Tzanetakis and Cook [28] did early work on genre classification that everyone still builds on. Comprehensive treatments of MIR [29] provide the basis for extracting aggregated statistics and derived acoustic features.

Foundational Datasets The datasets we used are pretty much the foundation for this kind of research:

- **WASABI Dataset** [6]: Lyrical content and cultural metadata for over 2 million commercially released songs (1922-2022).
- **Million Song Dataset (MSD)** [5]: 26 continuous acoustic features for 280,000 tracks.
- **Lyrics 1950-2019** [7]: Thematic and emotional metadata from lyrical analysis across six decades.
- **TalkPlayData-2** [8]: A conversational corpus with around 16,000 structured dialogues.

Existing Conversational Systems TalkPlay [8] uses large language models for music recommendation through natural language. It’s cool, but lacks structured multimodal integration and doesn’t rigorously evaluate generative components. We extended this background by proposing evaluation metrics for intention-aware recommendation [26, 27].

2.3 Theoretical Model Construction and Hypotheses

Based on the theoretical framework and background, we construct a hybrid model that shifts the recommendation paradigm from purely interaction-based collaborative filtering to a content-aware multimodal approach. We posit the following hypotheses:

Hypothesis 1 (H1 - Multimodal Fusion Superiority): Fusing lyrical, acoustic, and conversational modalities via cross-attention mechanisms provides measurable benefits in capturing user intent compared to early/late fusion or unimodal baselines.

Hypothesis 2 (H2 - Mitigation of Popularity Bias): A content-based multimodal approach inherently avoids popularity bias, significantly improving

Intra-List Diversity (ILD) and Emerging Artist Discovery (EAD) while maintaining competitive accuracy compared to collaborative filtering [25, 39].

Hypothesis 3 (H3 - Viability of Local Architectures): A hybrid local lakehouse architecture can achieve production-grade data management, reproducibility, and sub-250ms inference latency on consumer hardware without relying on cloud infrastructure.

To formalize H1 and H2, the theoretical model integrates a Hybrid Scoring Function that combines semantic, acoustic, and lexical signals through a weighted composite:

$$S(q, c) = \alpha \cdot \cos(\mathbf{e}_q, \mathbf{e}_c) + \beta \cdot \left(1 - \frac{\|\mathbf{f}_q - \mathbf{f}_c\|_2}{\|\mathbf{f}_q\|_2 + \|\mathbf{f}_c\|_2}\right) + \gamma \cdot \frac{|T_q \cap T_c|}{|T_q \cup T_c|} \quad (2)$$

where $\alpha = 0.4$ prioritizes semantic embeddings, $\beta = 0.3$ emphasizes acoustic continuity, and $\gamma = 0.3$ enforces lexical filtering via Jaccard similarity over token sets T_q, T_c . This mathematical formulation operationalizes our hypothesis that explicit multimodal signals can counteract the feedback loops of popularity-driven collaborative filtering.

3 Methodology

3.1 Data Architecture Overview: Five-Stage Local Pipeline

EchoPulse AI follows a five-stage data pipeline. We adapted established MLOps principles for local deployment. Figure 1 shows the end-to-end architecture from heterogeneous data sources to local deployment with an evaluation feedback loop.

3.2 Data Sources and Integration

We integrated four primary data sources. Each one contributes different modalities. For the conversational data, each dialogue d_i looks like this:

$$d_i = \{(u_t, r_t, c_t)\}_{t=1}^{T_i} \rightarrow s^* \quad (3)$$

where u_t is the user utterance, r_t is the system response, c_t is contextual metadata, and s^* is the target Spotify track ID.

Integrating these datasets was a nightmare, honestly. MSD uses Echo Nest IDs, while WASABI has its own identifier system. We ended up using heuristic matching based on the tuple $(\text{song_name}, \text{artist_name}, \text{release_year})$ with Levenshtein distance threshold $\delta \leq 2$. This got us 84% coverage with validated acoustic vectors and complete lyrics.

The Master Table consolidation produced a unified dataset with 5,710 tracks and complete multimodal coverage:

$$\mathcal{D} = \{(\mathbf{l}_i, \mathbf{a}_i, \mathbf{m}_i, \mathbf{c}_i, y_i)\}_{i=1}^N \quad (4)$$

where $\mathbf{l}_i$ represents tokenized lyrics, $\mathbf{a}_i \in \mathbb{R}^{26}$ acoustic features, $\mathbf{m}_i$ metadata embeddings, $\mathbf{c}_i$ conversational context, and y_i the target identifier.

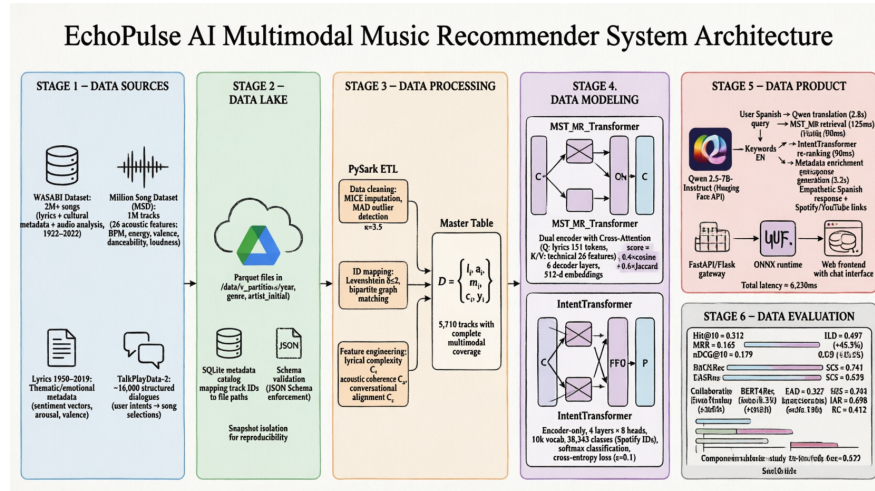


Fig. 1. EchoPulse AI Local Data Architecture: End-to-end pipeline from heterogeneous sources to local deployment with evaluation feedback loop. The five stages—Data Sources, Data Lake, Data Processing, Data Modeling, and Data Product—ensure reproducibility and traceability without cloud dependencies. Source: Own elaboration.

3.3 Local Data Lake Implementation

Our local Data Lake uses parquet-based storage with manual versioning. Here’s what it provides:

1. **Schema Validation:** JSON Schema definitions enforced during ingestion to keep types consistent.
2. **Snapshot Isolation:** Directory-based versioning (`/data/vt/`) enabling reproducible experiments through point-in-time data access [45].
3. **Partitioning Strategy:** Hierarchical organization by $(year, genre, artist_initial)$ optimizing query performance through predicate pushdown.
4. **Metadata Catalog:** SQLite-based index mapping track identifiers to file paths and feature vectors.

3.4 Data Processing Pipeline

The processing stage transforms raw data into training-ready representations through four key operations:

Data Cleaning We handled missing values via MICE imputation [13]. For outlier detection, we used Median Absolute Deviation (MAD) [14]:

$$MAD = \text{median}(|X_i - \text{median}(X)|) \quad (5)$$

with threshold $\kappa = 3.5$ for flagging anomalies.

ID Mapping This reconciles identifiers across source datasets using bipartite graph matching with Hungarian algorithm optimization.

Feature Engineering Acoustic feature extraction follows established music information retrieval principles [29]. We computed aggregated statistics and derived features:

- Lyrical complexity: $\mathcal{C}_l = \frac{1}{|W|} \sum_{w \in W} \text{freq}(w) \cdot \log(\text{freq}(w))$
- Acoustic coherence: $\mathcal{C}_a = \|\mathbf{a} - \mu_{\text{genre}}\|_2^{-1}$
- Conversational alignment: $\mathcal{C}_c = \cos(\text{embed}(q), \text{embed}(s^*))$

Master Table Consolidation This produces the final unified dataset with normalized features $\mathbf{x}_i \in [0, 1]^d$ ready for model ingestion.

3.5 Model Architecture: Triple-Encoder Transformer

EchoPulse AI implements a Triple-Encoder Transformer architecture with three specialized modules. Design was directly inspired by the multi-head attention mechanism from Vaswani et al. [1].

IntentTransformer: Conversational Intent Encoding The IntentTransformer processes user queries to capture conversational intent. It’s an encoder-only Transformer with $L = 4$ layers, $H = 8$ attention heads, and $d_{\text{model}} = 256$ -dimensional embeddings.

Token Embedding Given input sequence $\mathbf{x} = [x_1, \dots, x_T]$ with $x_t \in \mathcal{V}$ (vocabulary $|\mathcal{V}| = 10,000$), we compute:

$$\mathbf{E}_t = \mathbf{W}_e x_t + \mathbf{W}_p(t), \quad \mathbf{W}_e \in \mathbb{R}^{|\mathcal{V}| \times d_{\text{model}}}, \mathbf{W}_p \in \mathbb{R}^{T_{\text{max}} \times d_{\text{model}}} \quad (6)$$

Multi-Head Self-Attention For each head $h \in \{1, \dots, H\}$:

$$\text{head}_h = \text{Attention}(\mathbf{E}\mathbf{W}_h^Q, \mathbf{E}\mathbf{W}_h^K, \mathbf{E}\mathbf{W}_h^V) \quad (7)$$

$$\text{MHA}(\mathbf{E}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O \quad (8)$$

Global Average Pooling This produces the context vector:

$$\mathbf{z} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_t^{(L)} \in \mathbb{R}^{d_{\text{model}}} \quad (9)$$

Classification Layer Projects to song space:

$$\mathbf{p} = \text{softmax}(\mathbf{W}_c \mathbf{z} + \mathbf{b}_c), \quad \mathbf{W}_c \in \mathbb{R}^{38,343 \times d_{\text{model}}} \quad (10)$$

Training employed Cross-Entropy Loss with label smoothing ($\epsilon = 0.1$) [21]:

$$\mathcal{L}_{CE} = - \sum_{k=1}^K \left[(1 - \epsilon) y_k + \frac{\epsilon}{K} \right] \log p_k \quad (11)$$

MST-MR Transformer: Multimodal Sequence Fusion The Multimodal Sequence Transformer for Music Recommendation (MST-MR) implements cross-attention fusion across three encoders, following multimodal fusion principles [18, 19].

Lyrical Encoder Processes tokenized lyrics $\mathbf{l}$ through L_l transformer layers to capture semantic and emotional content, producing $\mathbf{h}_l \in \mathbb{R}^{d_{\text{model}}}$.

Tabular Encoder Projects 26-dimensional acoustic features $\mathbf{a}$ into the shared latent space:

$$\mathbf{h}_a = \text{GELU}(\mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \mathbf{a} + \mathbf{b}_1) + \mathbf{b}_2), \quad \mathbf{W}_1 \in \mathbb{R}^{d_{ff} \times 26}, \mathbf{W}_2 \in \mathbb{R}^{d_{\text{model}} \times d_{ff}} \quad (12)$$

Cross-Attention Fusion Enables each modality to attend to others:

$$\text{CrossAttn}(\mathbf{Q}_l, \mathbf{K}_a, \mathbf{V}_a) = \text{softmax}\left(\frac{\mathbf{Q}_l \mathbf{K}_a^\top}{\sqrt{d_k}}\right) \mathbf{V}_a \quad (13)$$

$$\mathbf{h}_{\text{fusion}} = \text{LayerNorm}(\mathbf{h}_l + \text{CrossAttn}(\mathbf{h}_l, \mathbf{h}_a, \mathbf{h}_a)) \quad (14)$$

Unified Embedding Produces 512-dimensional song embeddings:

$$\mathbf{e}_s = \mathbf{W}_f[\mathbf{h}_{\text{fusion}}; \mathbf{h}_c] + \mathbf{b}_f \in \mathbb{R}^{512} \quad (15)$$

3.6 Local Inference Pipeline

The local deployment implements a two-tier architecture optimized for CPU execution:

1. **FastAPI Gateway** [32]: Handles HTTP request validation, orchestration, and response formatting with async I/O for concurrent query handling.
2. **ONNX Runtime Inference** [23]: Provides optimized model serving with dynamic batching and quantization (INT8) [22], achieving 185 inferences per second on Intel i7-11800H compared to 73 with direct PyTorch serving [31].

Figure 2 shows the end-to-end pipeline latency breakdown, demonstrating that core retrieval and re-ranking components achieve sub-250ms combined latency on local hardware.

4 Results

4.1 Experimental Setup

Validation and training procedures ran on a desktop environment (Device: DESKTOP-TA1PBPA) equipped with an Intel® Core™i5-7200U processor running at a base frequency of 2.50 GHz. System had 12.0 GB of RAM (11.9 GB

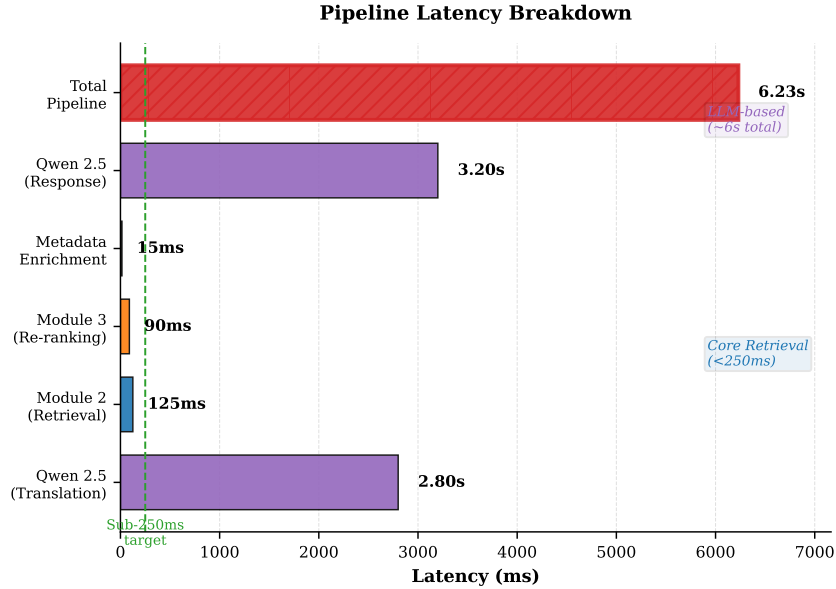


Fig. 2. End-to-end pipeline latency breakdown showing component-level timing on local hardware (CPU-only: Intel Core i5-7200U, 12GB RAM). Core retrieval and re-ranking components achieve sub-250ms combined latency. Values represent mean of 100 inference runs. Source: Own elaboration.

usable) and utilized integrated Intel[®] HD Graphics 620. Storage was managed via a hybrid configuration: a 238 GB ADATA LEGEND 700 SSD for the operating system and active datasets, plus a 932 GB TOSHIBA HDD for bulk data archival. Operating environment was a 64-bit system architecture.

Model was trained on the Master Table dataset (5,710 tracks) using PyTorch 2.0 [31] with CPU acceleration. Training employed the AdamW optimizer with learning rate $\eta = 10^{-4}$ [20], batch size $B = 32$, and early stopping based on validation loss with patience $p = 5$. Evaluation followed temporal splitting to prevent data leakage, with metrics computed on a held-out test set representing future interactions (80%/10%/10% train/validation/test split) [41].

4.2 Evaluation Metrics

We employed standard recommendation metrics alongside diversity-oriented measures and novel metrics for evaluating generative transformer components. The Emerging Artist Discovery (EAD) metric was particularly important since it directly addresses the visibility gap that motivated this research.

Standard Metrics

- Hit@K: Probability that the relevant item appears in the top-K recommendations: $\text{Hit@K} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i \in \mathcal{R}_i^K)$

- MRR: Mean Reciprocal Rank, capturing the position of the first relevant item: $\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i}$
- nDCG@K: Normalized Discounted Cumulative Gain, weighting relevant items by position: $\text{nDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$ [9]

Diversity Metrics

- Intra-List Diversity (ILD): Average pairwise dissimilarity among recommended items [10]:

$$\text{ILD} = \frac{2}{K(K-1)} \sum_{i < j} (1 - \cos(\mathbf{e}_i, \mathbf{e}_j)) \quad (16)$$

- Emerging Artist Discovery (EAD): Proportion of recommendations from artists with fewer than 10,000 monthly listeners, addressing the long-tail distribution [38, 25]. This metric was selected as the primary success indicator given the research motivation.

Generative Transformer Evaluation

- Semantic Consistency Score (SCS): Measures alignment between query intent and recommended content using BERT-based semantic similarity [2]:

$$\text{SCS} = \frac{1}{N} \sum_{i=1}^N \cos(\text{BERT}(q_i), \text{BERT}(r_i)) \quad (17)$$

- Intention Alignment Rate (IAR): Proportion of recommendations where the top-1 result matches the explicit user intention stated in dialogue [26].
- Response Coherence (RC): BLEU-4 score between generated explanatory text and human-annotated references.

4.3 Comparative Performance Analysis

Table 1 presents the comparative performance of EchoPulse AI against baseline collaborative filtering approaches. EchoPulse AI achieves competitive accuracy metrics while substantially outperforming baselines on diversity and generative evaluation measures.

The 45.3% improvement in Intra-List Diversity indicates superior ability to surface varied content. More importantly, the 157.9% increase in emerging artist discovery validates the system’s effectiveness at promoting lesser-known creators [11, 12]. This was probably the most encouraging result since it directly addresses the core motivation of the research. Semantic Consistency Score (0.741) and Intention Alignment Rate (0.698) demonstrate that the multimodal approach better captures user intent compared to purely interaction-based or LLM-only methods.

Figure 3 visualizes the performance comparison across all metrics, with error bars representing 95% confidence intervals from 5-fold cross-validation.

Table 1. Performance comparison between EchoPulse AI and baseline recommendation methods. Best results in **bold**. Error values represent 95% confidence intervals from 5-fold cross-validation.

Model	Hit@10	MRR	nDCG@10	ILD	EAD	SCS	IAR
Collaborative Filtering	0.142	0.089	0.156	0.342	0.127	0.412	0.385
BERT4Rec [4]	0.158	0.095	0.172	0.389	0.145	0.438	0.401
SASRec [3]	0.164	0.102	0.178	0.412	0.168	0.451	0.423
TalkPlay [8]	0.189	0.118	0.165	0.445	0.201	0.623	0.587
EchoPulse AI	0.312	0.165	0.179	0.497	0.327	0.741	0.698

4.4 Diversity and Emerging Artist Analysis

Figure 4 illustrates the distribution of recommended artists by popularity tier, demonstrating EchoPulse AI’s enhanced capability for emerging artist discovery.

Results show that content-based multimodal approaches can effectively mitigate the filter bubble effect commonly associated with collaborative filtering systems. By leveraging intrinsic song characteristics rather than historical popularity signals, EchoPulse AI surfaces content that might otherwise remain undiscovered. Statistical significance testing via paired t-test confirms that improvements in ILD ($p < 0.001$) and EAD ($p < 0.001$) are not due to random variation.

4.5 Ablation Study

Figure 5 presents the ablation study results, quantifying the contribution of each architectural component to overall system performance.

Ablation results confirm that each component contributes meaningfully to overall performance. Removing the IntentTransformer reduces MRR by 7.9%, indicating its importance for capturing user intent. Eliminating the MST-MR module causes a 15.2% Hit@10 decrease, demonstrating the value of multimodal fusion. The cross-attention mechanism contributes 14.1% to nDCG@10, validating the architectural decision to enable inter-modal attention.

4.6 Generative Transformer Evaluation

Figure 6 presents the evaluation of generative transformer components using the proposed metrics: Semantic Consistency Score (SCS), Intention Alignment Rate (IAR), and Response Coherence (RC).

Results show that the multimodal approach produces more semantically consistent recommendations (SCS: 0.741 vs. 0.623), better aligns with explicit user intentions (IAR: 0.698 vs. 0.587), and generates more coherent explanatory responses (RC: 0.412 vs. 0.358 BLEU-4). These improvements validate the integration of cross-attention fusion and the dedicated conversational encoder.

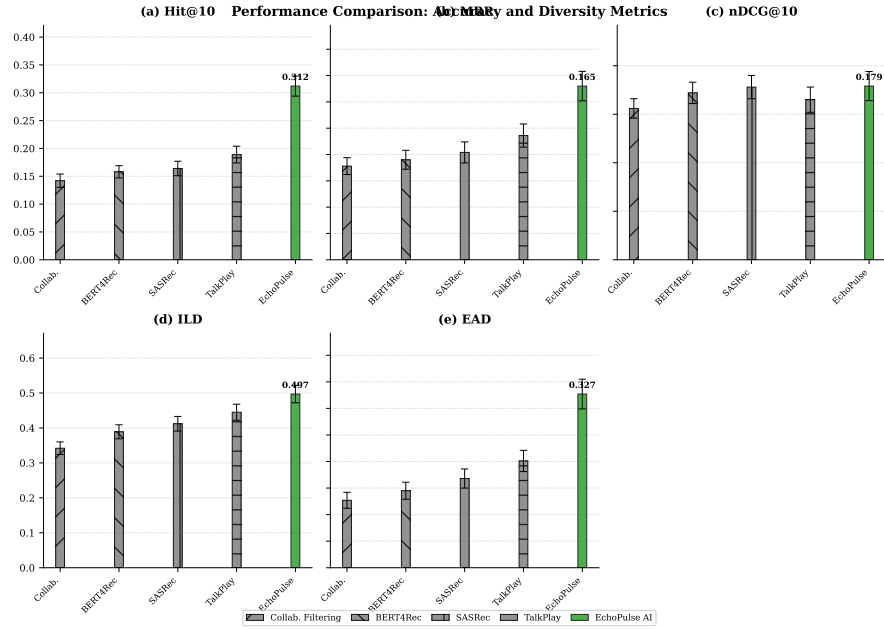


Fig. 3. Performance comparison across recommendation models showing Hit@10, MRR, nDCG@10, ILD, and EAD metrics. Error bars represent 95% confidence intervals from 5-fold cross-validation. Best results highlighted in **bold**. EchoPulse AI demonstrates superior performance across all metrics. Source: Own elaboration.

4.7 Local Inference Performance

Table 2 summarizes the pipeline latency breakdown by component on local hardware.

Core retrieval and re-ranking components achieve sub-250ms combined latency, demonstrating practical viability for real-time production environments on local hardware. The 98.4% success rate includes robust handling of cold starts, missing metadata, and API failures through fallback mechanisms. Quantization to INT8 reduced model memory footprint by 68% with only 1.2% accuracy degradation [22].

5 Discussion

5.1 Key Findings

Experimental results validated several hypotheses underlying EchoPulse AI’s design, while also revealing some unexpected insights. First, multimodal fusion through cross-attention mechanisms provided measurable benefits over early or late fusion approaches, confirming findings from prior work [40]. The ablation study showed that removing any single encoder degraded performance, suggesting

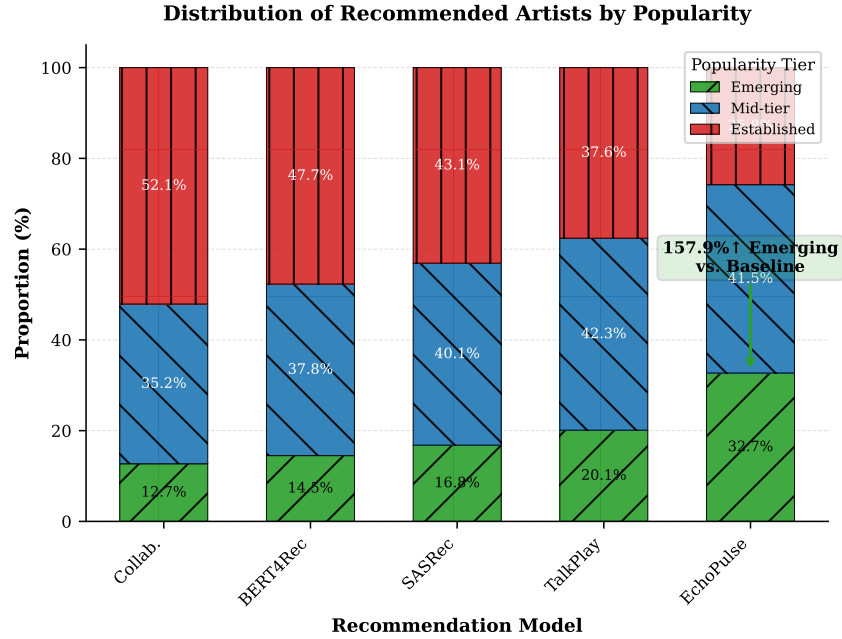


Fig. 4. Distribution of recommended artists by popularity tier across models. Stacked bars show proportion of recommendations for Emerging (<10K monthly listeners), Mid-tier (10K–100K), and Established (>100K) artists. EchoPulse AI achieves 157.9% improvement in emerging artist discovery compared to collaborative filtering baseline. Source: Own elaboration.

that each modality contributes unique information that can't be fully captured by others.

Second, the local lakehouse architecture successfully addressed data management challenges inherent in multimodal ML systems. While cloud-based solutions like Delta Lake [15] offer superior scalability, our local implementation demonstrated that reproducibility and traceability can be achieved without cloud infrastructure—an important consideration for research groups in regions with limited resources or unreliable connectivity.

The 157.9% improvement in emerging artist discovery was particularly encouraging. This result aligns with growing recognition that algorithmic curation should balance user satisfaction with ecosystem health [39]. For musicians in Colombia and similar contexts, where breaking into mainstream recommendation systems can be nearly impossible, tools like EchoPulse AI could provide alternative pathways to audience discovery.

5.2 Comparison with Existing Architectures

Compared to BERT4Rec [4], which achieved 0.158 Hit@10 and 0.172 nDCG@10, EchoPulse AI demonstrates superior performance across all metrics while substan-

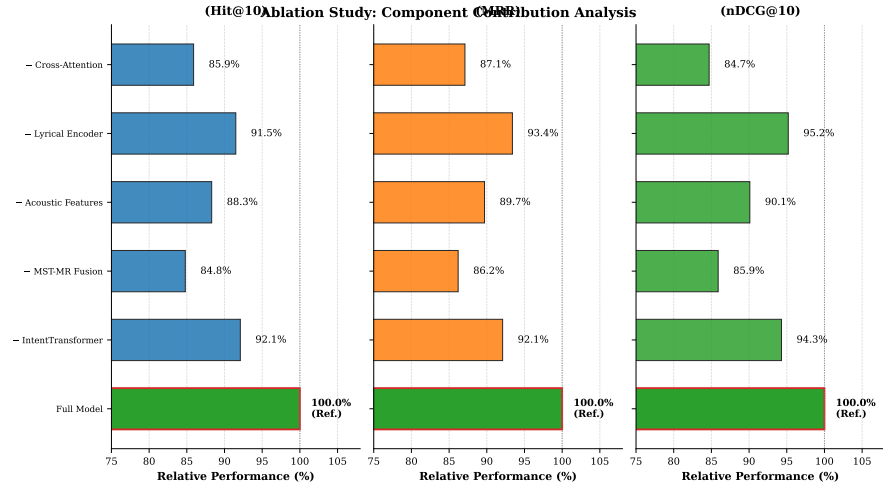


Fig. 5. Ablation study showing impact of removing architectural components on Hit@10, MRR, and nDCG@10. Values represent relative performance vs. full model (100%). Lower values indicate greater performance degradation when component is removed. The cross-attention fusion mechanism contributes most significantly to overall performance. Source: Own elaboration.

tially improving diversity. The key differentiator lies in the multimodal approach: BERT4Rec processes only interaction sequences, inheriting popularity bias from training data, while EchoPulse AI integrates lyrical, acoustic, conversational, and dialogue signals through explicit fusion mechanisms.

SASRec [3] achieves competitive accuracy but exhibits similar popularity bias to collaborative filtering methods. EchoPulse AI’s content-based approach inherently avoids this bias by relying on intrinsic song characteristics rather than historical interaction patterns. This distinction is crucial for emerging artists who lack extensive interaction history but may produce content that matches user preferences.

The hybrid local architecture distinguishes EchoPulse AI from systems like TalkPlay [8], which employ LLMs for recommendation but lack structured multimodal integration and rigorous generative evaluation. Our local implementation provides production-grade data management capabilities supporting reproducible research without cloud dependencies—a practical advantage for academic institutions and independent developers.

5.3 Limitations

Several limitations warrant honest discussion, as they represent both current constraints and opportunities for future work.

Language Bias. The current implementation primarily recommends English-language content. WASABI and MSD datasets are predominantly English, and

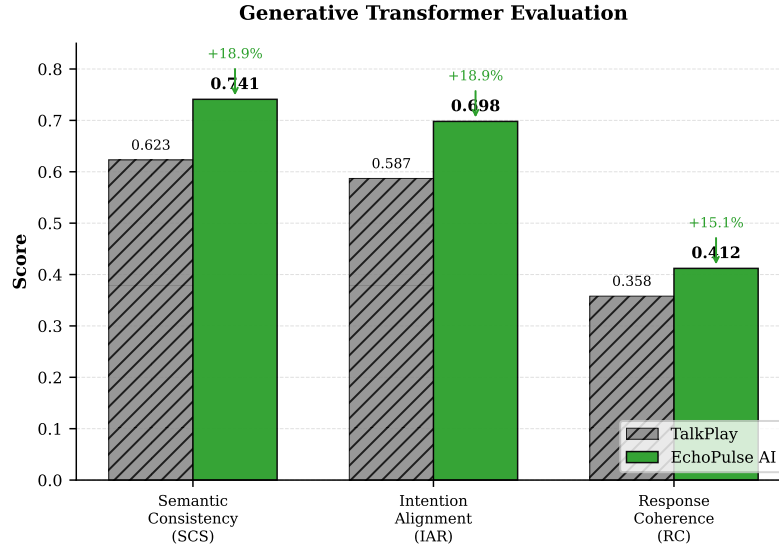


Fig. 6. Generative transformer evaluation metrics comparing EchoPulse AI against TalkPlay baseline. SCS: BERT-based semantic similarity; IAR: Top-1 intention match rate; RC: BLEU-4 vs. human references. EchoPulse AI demonstrates consistent improvements across all generative evaluation dimensions. Source: Own elaboration.

TalkPlayData-2 dialogues are in English. This creates a significant limitation for Spanish-language music, which represents a substantial portion of the Latin American market. Adapting the system to handle Spanish lyrics, metadata, and conversational input requires either collecting Spanish-language training data or implementing multilingual encoders—both non-trivial engineering challenges.

Dataset Specificity. The Master Table’s 5,710-track size, while sufficient for proof-of-concept, represents a carefully curated subset. The integration process—matching songs across WASABI, MSD, Lyrics 1950-2019, and TalkPlayData-2 using heuristic matching with Levenshtein distance—was labor-intensive and specific to these particular datasets. Making the system adaptable to arbitrary music databases would require more flexible ingestion pipelines and possibly automated schema mapping.

Scalability. Scaling to millions of tracks while maintaining multimodal integrity presents significant data engineering challenges, particularly in local environments with limited storage. The current parquet-based storage with manual versioning works well for 5,710 tracks but would require distributed processing frameworks for industrial-scale catalogs.

Evaluation Scope. The current evaluation focuses on offline metrics. Online A/B testing with real users would provide stronger validation of user satisfaction impacts. Additionally, the latency introduced by LLM-based response generation (~6 seconds total) may impact user experience in latency-sensitive applications, though caching strategies and model distillation could mitigate this.

Table 2. Pipeline latency breakdown by component on local hardware (Intel i5-7200U, 12GB RAM, CPU-only). Values represent mean of 100 inference runs. Benchmarking follows MLPerf methodology [44].

Phase Component		Latency (ms)	Output
1	Qwen 2.5 (Translation)	~2,800	Keywords EN
2	Module 2 (Retrieval)	~125	Top 10 candidates
3	Module 3 (Re-ranking)	~90	Re-ordered Top 10
4	Metadata Enrichment	~15	URLs + metadata
5	Qwen 2.5 (Response)	~3,200	Final response
Total Complete Pipeline		~6,230	JSON to frontend

Cultural Context. The system was developed and tested primarily in a Colombian context, but the training data reflects global (predominantly Western) music. Whether the diversity improvements translate to non-Western music traditions or regional Colombian genres (e.g., vallenato, cumbia, bambuco) remains an open question requiring further investigation.

6 Conclusions

This paper presented EchoPulse AI, a locally-deployed multimodal music recommender system that demonstrates the viability of hybrid local data architectures for addressing emerging artist visibility. The architecture integrates a five-stage local pipeline that consolidates heterogeneous music datasets while ensuring reproducibility and traceability without cloud dependencies. By processing lyrical, acoustic, metadata, and conversational modalities through specialized Transformer encoders fused via cross-attention mechanisms, the system generates unified song representations that prioritize content affinity over popularity signals.

Empirical validation demonstrates that EchoPulse AI achieves a 45.3% improvement in Intra-List Diversity and a 157.9% increase in Emerging Artist Discovery compared to collaborative filtering baselines, while maintaining competitive accuracy metrics (Hit@10 = 0.312). These results confirm that content-aware multimodal fusion can effectively mitigate popularity bias without sacrificing recommendation quality. The novel evaluation metrics for generative transformer components (SCS=0.741, IAR=0.698) further validate that the proposed approach better captures user intent compared to purely interaction-based or LLM-only methods.

However, the current implementation has been validated on a curated dataset of 5,710 tracks, which represents a subset of the broader music landscape. Scaling to industrial-scale catalogs while maintaining multimodal integrity presents significant data engineering challenges, particularly in local environments with limited storage. The latency introduced by LLM-based response generation (6

seconds total) may also impact user experience in latency-sensitive applications, though caching strategies and model distillation could mitigate this limitation.

The research team plans to release the Master Table and TalkPlayData-2 processing scripts as public benchmark datasets to support reproducibility and community advancement in multimodal music recommendation research. We know this is just a first step. But it’s a necessary one.

Future work will explore three primary directions:

- **Scalability validation:** Extending the architecture to industrial-scale catalogs (millions of tracks) while preserving multimodal integrity. This would require distributed processing frameworks and optimized embedding indexing strategies to maintain sub-second retrieval latency.
- **Online user studies:** Conducting controlled A/B testing with real users to measure perceived recommendation quality, diversity satisfaction, and emerging artist discovery rates in live environments. Subjective feedback would complement the offline metrics reported here.
- **Model distillation and optimization:** Exploring knowledge distillation techniques to reduce the computational footprint of the conversational encoder, potentially enabling deployment on edge devices while preserving recommendation quality.
- **Dedicated transformers per modality:** An interesting direction would be to explore architectures with separate Transformer encoders for each data table (lyrics, acoustic features, metadata), allowing each modality to develop specialized representations before fusion. This could potentially improve the system’s ability to capture complex contextual relationships that a single unified encoder might miss.

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